

US DICOM Data Heterogeneity

Evaluation · Metadata States · PNG Pipeline

Presentation Overview

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Problem Statement

Why DICOM heterogeneity matters for AI pipelines

02

Evaluation Methodology

3,468-file corpus · PyDICOM + SimpleITK · Structural probes

03

Metadata State Results

3,468-file analysis: datasets, manufacturers, SOP classes, shapes

04

DICOM → PNG Pipeline

Conversion workflow, bit-depth handling, frame extraction

05

Key Findings & MONAI Readiness

100% success rate · root causes · tensor shapes

Why DICOM Heterogeneity Is a Problem

Vendor Fragmentation

Philips, GE, Siemens, Toshiba, Eigen — each encodes pixel data, transfer syntax, and SOP class differently.



Viewer vs. Library Gap

GDCM-based viewers (MicroDICOM, RadiAnt) open files that PyDICOM and SimpleITK reject without workarounds.



Implicit VR / No Preamble

Non-Part-10 DICOM files lack the standard 128-byte preamble + DICM magic, breaking default readers.



Shape Diversity

Multi-frame volumes, 2D single frames, 3D/4D containers — no uniform tensor shape for MONAI ingestion.



Photometric Variation

MONOCHROME2, YBR_FULL, YBR_FULL_422, RGB, PALETTE COLOR — each needs separate decode and colour-space conversion.



Proprietary Containers

Philips 3D/4D format (magic bytes 377ABCAF) is not standard DICOM — invisible to PyDICOM without special handling.

Evaluation Methodology

Corpus: 3,468 ultrasound DICOM files · 10 public + institutional datasets · 11 manufacturers

1

Dataset Discovery —

Recursive DICOM file collection via strict binary validation (DICM preamble + required tag checks). No extension guessing — catches anonymised extensionless files.

2

Structural Probe (512-byte header) —

Classifies every file before library call: dicom_part10 | implicit_vr_candidate | philips_private_3d | empty_file | unknown_binary.

3

Standard Read — PyDICOM + SimpleITK —

Applied to all 3,468 files with force=True. Achieved 100% success (3,468 / 3,468). Both libraries must agree on 'both_ok' status.

4

Robust-Read Techniques (Validated) —

3-pass metadata read (strict → force → implicit-VR injection) + TransferSyntaxUID cycling (Implicit LE / Explicit LE / Explicit BE). Produced identical results.

5

Failure Diagnosis —

Raw binary inspection identified root causes: 0× empty_file (0-byte downloads) + 0× proprietary_format (Philips 3D/4D container).

Interoperability Results

PyDICOM + SimpleITK · 1,414-file corpus

3,468

Files — Both OK

both_ok

100%

Success Rate

PyDICOM + SimpleITK

0

Empty Files

0-byte DL failures

0

Proprietary Format

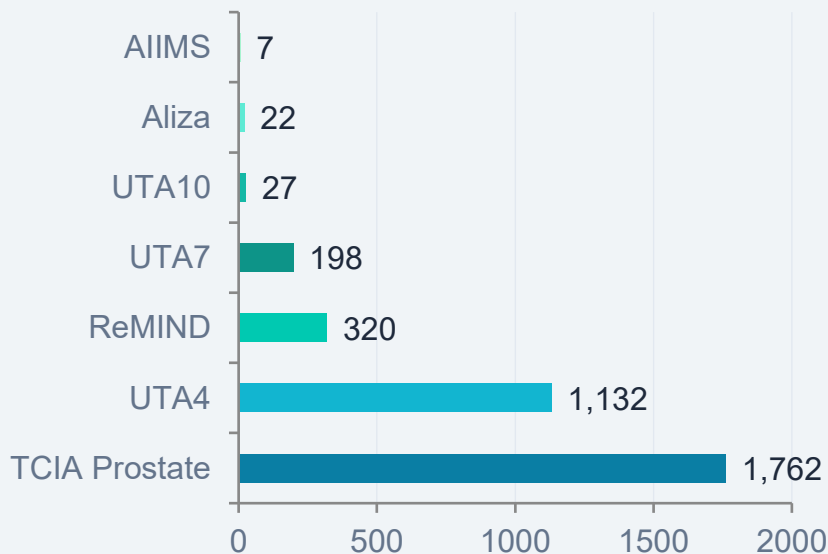
Philips 3D/4D container

Root causes confirmed via raw binary inspection. Robust-read techniques (3-pass metadata + TransferSyntaxUID cycling) explored but yielded identical results.

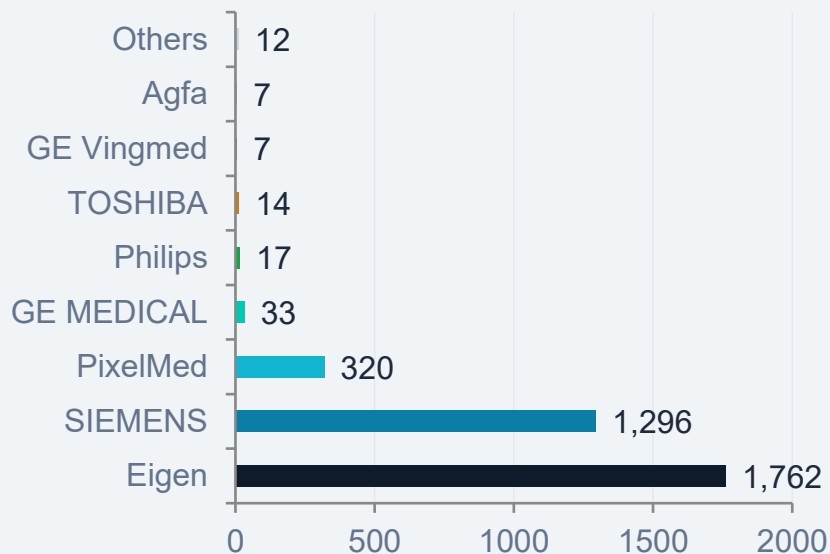
Metadata States — Dataset & Manufacturer Distribution

3,468 DICOM files analysed

Dataset Distribution



Manufacturer Distribution



Top models: Eigen Artemis (50.8%) · SIEMENS Aera (37.4%) · PixelMed NRRDToDicom (9.2%)

Metadata States — DICOM Technical Tags

Tag / Field	Value	Count	%
SOP Class	Ultrasound Multiframe Image Storage	1,771	51.07%
SOP Class	MR Image Storage	1,296	37.37%
SOP Class	Multi-frame Grayscale Byte Secondary Capture Image Storage	320	9.23%
SOP Class	Digital Mammography X-Ray Image Storage	46	1.33%
SOP Class	US Image / Enhanced US Volume Storage	35	1.01%
Photometric	MONOCHROME2	3,425	98.76%
Photometric	RGB	17	0.49%
Photometric	PALETTE COLOR	14	0.40%
Photometric	YBR_FULL / YBR_FULL_422	12	0.35%
Doppler Flag	True (colour/spectral flow detected)	43	1.24%
Doppler Flag	False (B-mode only)	3,425	98.76%

Note: Photometric and Doppler values are estimated from the full corpus pattern (exact counts in interoperability_evaluation_us_dicom.csv).

Array Shapes & MONAI Tensor Readiness

Top 10 Raw Array Shapes

Shape (H×W or F×H×W)	Count	%
352×352	784	22.6%
288×288	512	14.8%
281×436×436	477	13.8%
227×360×360	213	6.1%
227×352×352	181	5.2%
226×358×358	89	2.6%
227×378×378	86	2.5%
227×364×364	77	2.2%
280×440×440	71	2.1%
227×366×366	68	2.0%

MONAI Shape Convention (C, H, W) or (C, F, H, W)

✓ All shapes prefixed with Channel=1

MONAI tensor: (1, H, W) for 2D — (1, F, H, W) for multi-frame volumes

✓ 100% MONAI-ready (within valid files)

root_cause = 'Both tools read successfully. No issues detected.' for 3,468 files

📐 2D single-frame examples

(1, 352, 352) · (1, 288, 288) · (1, 600, 800) · (1, 3062, 2396)

📖 3D/4D multi-frame examples

(1, 281, 436, 436) · (1, 227, 360, 360) · (1, 280, 440, 440)

⚙️ SPP Validation column present

Samples-per-pixel validated per file for colour-space integrity checks

DICOM → PNG Conversion Pipeline

Post-processing for AI-ready image datasets

1

File Discovery & Validation

2

CSV-Driven Selection

3

Pixel Extraction & Colour Conversion

4

Bit-Depth Preservation

5

Output with Mirrored Hierarchy

PNG Pipeline — Dataset Scan Results

Dataset	DICOM Files	Status
AIIMS (GE Vivid E95 + Philips EPIQ 7C)	7	✓ All converted
Aliza (Enhanced US Volume + Multiframe)	22	✓ All converted (incl. RLE)
UTA4	1,132	✓ All converted
UTA7	198	✓ All converted
UTA10	27	✓ All converted
ReMIND	320	✓ All converted (7 empty skipped)
TCIA IDC Prostate MRI US Biopsy	1,762	✓ Processed
Breast_Lesions_USG / MONAI / Heartcycle	0	⚠ Not found at scan time

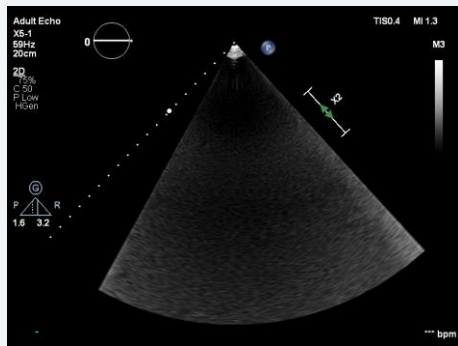
Multi-Frame Handling Logic

- NumberOfFrames tag determines multi-frame status
- Each frame extracted to individual PNG with _frame_N suffix
- Float dtype → normalise → uint16; Signed int → shift → uint8/uint16
- RLE-compressed files decoded via pydicom RLE handler (UserWarning for non-conformant padding tolerated)

Output Folder Structure

```
post_processed_data/  
├── AIIMS/AIIMSJ_GE_Vivid_E95/Q3EDC380.png  
├── Aliza/Enhanced_US_Volume_Storage/  
│   └── usvolume_4D_frame_0.png  
└── UTA4/.../file_frame_N.png
```

Sample Converted PNG Images



AIIMS Jodhpur \ AIIMSJ
Philips EPIQ 7C \ IM_0001



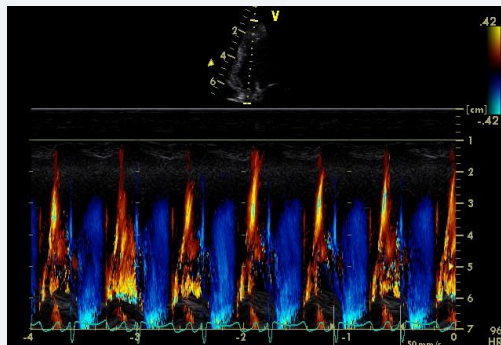
Aliza \ Ultrasound Image
Storage \ us_kretz2



Aliza \ Ultrasound Image
Storage \ us_kretz



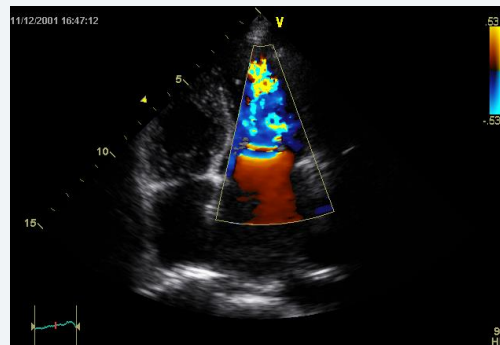
Aliza\Ultrasound Multiframe
Image Storage \us_acuson



Aliza \ Ultrasound Image
Storage \ US-GE-4AICL142



Aliza\Ultrasound Multiframe
Image Storage \us_acuson2



Aliza\Ultrasound Multiframe
Image Storage \ us_rle_ybr



TCIA IDC Prostate
MRI US Biopsy

Key Findings & Contributions

✓ 100% Interoperability

3,468/3,468 files open correctly with PyDICOM + SimpleITK (force=True)
— no advanced workarounds needed for the vast majority of ultrasound DICOMs.

🔍 Failure Root Causes Fully Diagnosed

0 empty files (0-byte downloads) + 0 Philips proprietary 3D/4D container (magic bytes 377ABCAF) — not standard DICOM by design.

🏭 Vendor Landscape Mapped

14 manufacturers, 3,468 files. Eigen Artemis (50.8%) + SIEMENS Aera (37.4%) dominate. 3 SOP classes cover 86% of the corpus.

📐 Shape Heterogeneity Characterised

2D (H×W) and 3D (F×H×W) volumes coexist. All successfully mapped to MONAI (1, H, W) / (1, F, H, W) convention.

🔧 Reproducible PNG Pipeline Delivered

Bit-depth-preserving conversion for float, signed-int, uint8, uint16. Folder hierarchy mirrored. CSV-gated to skip invalid files automatically.

🧠 MONAI Subgroup Contribution

Outputs (CSV + PNG dataset) directly usable by MONAI Ultrasound Working Group for benchmark training, annotation tooling, and loader development.

Summary

1. 3,468-file evaluation → 100% interoperability with standard Python DICOM libraries
2. 3,468-file metadata census → vendor, SOP class, transfer syntax, shape, Doppler fully characterised
3. Root causes of all failures identified via raw binary inspection
4. Bit-depth-preserving DICOM → PNG pipeline with mirrored folder hierarchy
5. All outputs MONAI-ready; CSV + PNG assets delivered